

Cryptograph and cyber law

Assignment - 06

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Assignment topic:

Modes of Operation and RC5 - Block Diagram and Java Implementation and output.

Modes of operation:-

Block ciphers encrypt data in fixed size blocks. (64 or 128 bits). But in real applications, data is often much longer. So, we use Modes of Operations to securely process large data by using a block cipher repeatedly.

Common modes:-

- ECB - Electronic codebook
- CBC - Cipher Block chaining
- CFB - Cipher Feedback
- OFB - Output Feedback
- CTR - Counter.

Description:

ECB: Each block is encrypted independently. Not secure for patterns.

CBC: XORs each plaintext block with previous ciphertext block before encryption.

CFB: Converts block cipher into a self-synchronizing stream cipher.

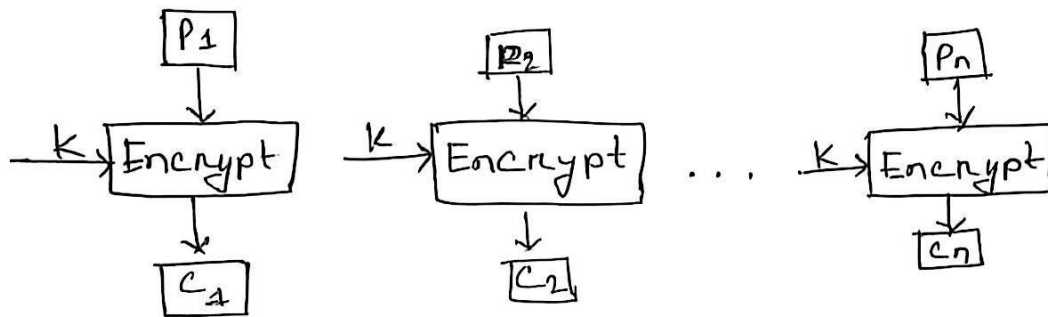
OFB: Turns block cipher into a synchronous stream cipher.

CTR: Uses a counter that gets encrypted and XORed with plaintext. Fast and parallelizable.

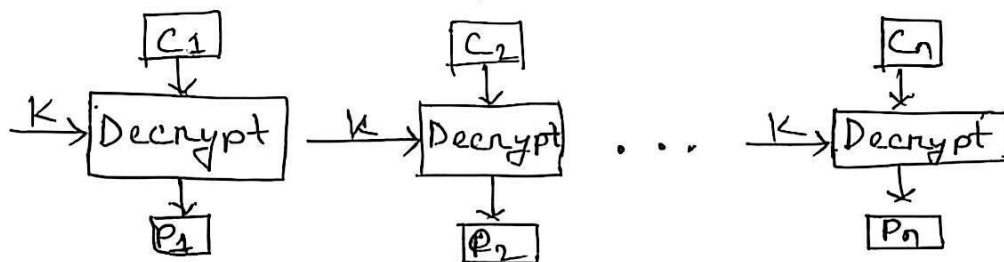
Note:- Among them, CBC and CTR are the most widely used due to their security and efficiency.

The procedure of ECB is illustrated below:

Encryption



Decryption



Advantages of ECB:

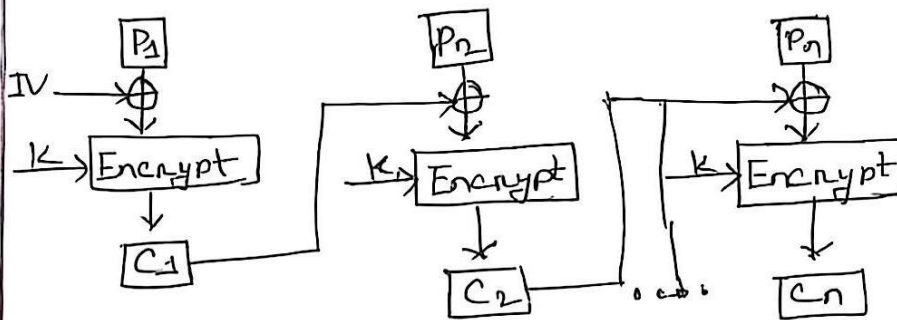
- Parallel ~~enery~~ encryption of blocks of bits is possible, thus it is a faster way of encryption.
- Simple way of the block cipher

Disadvantages of ECB:

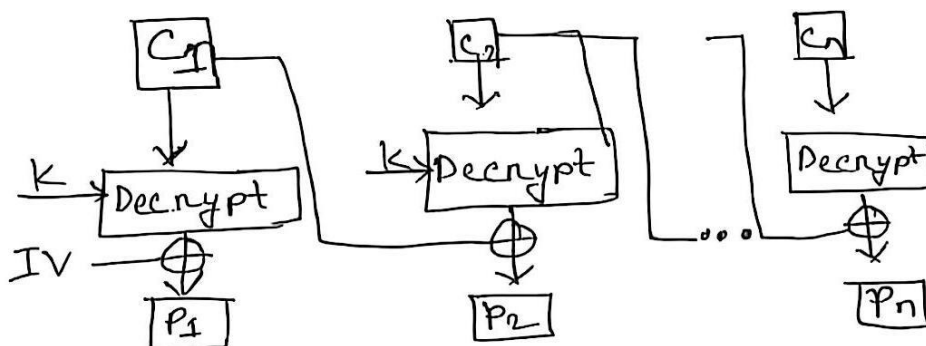
- Prone to cryptanalysis of block of bits is possible, thus it is a faster way of encryption
- Simple way of the block cipher.

CBC Block Diagram:

Encryption:



Decryption:



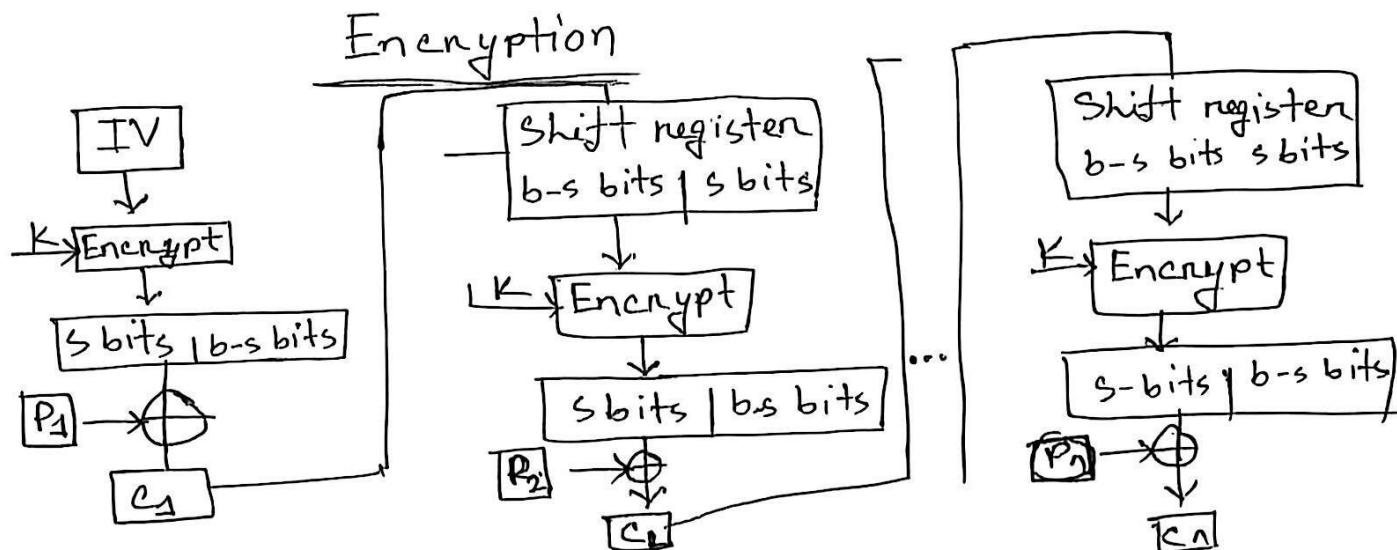
Advantage of CBC:

- CBC works well for input greater than b bits.
- CBC is a good authentication mechanism.
- Better resistive nature towards cryptanalysis than ECB.
- More secure than ECB as it hides patterns.

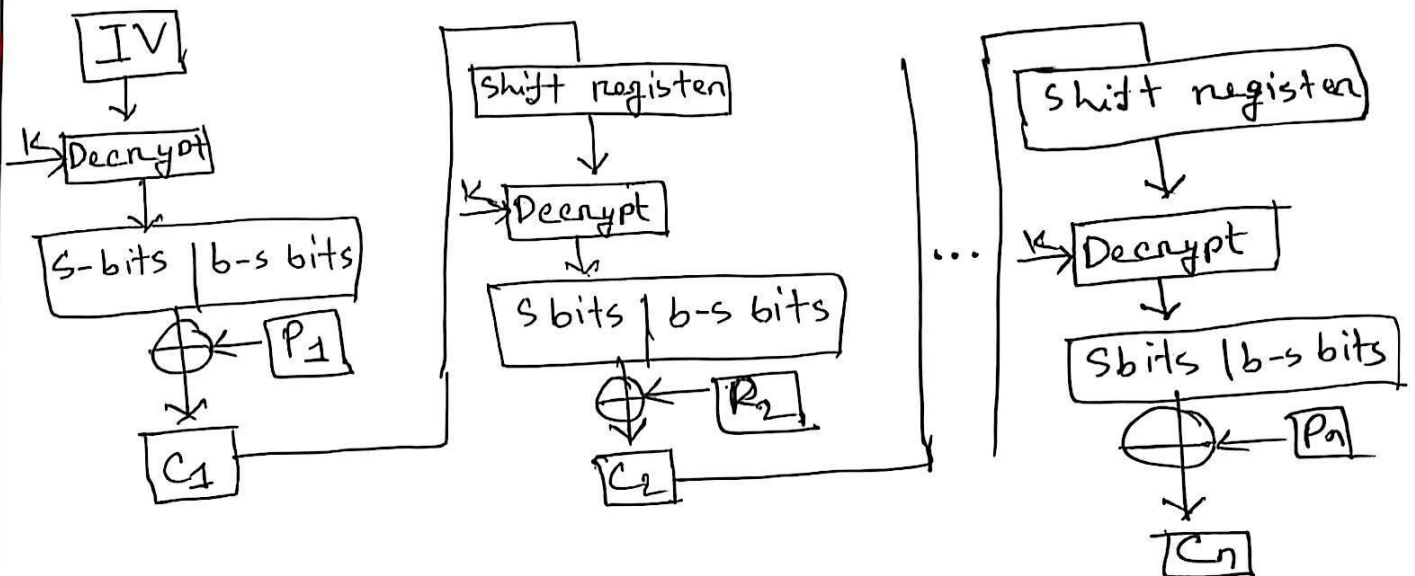
Disadvantages:

- Requires the previous ciphertext block for encryption and decryption.

CFB Block Diagram:



Decryption



Advantage of CFB:

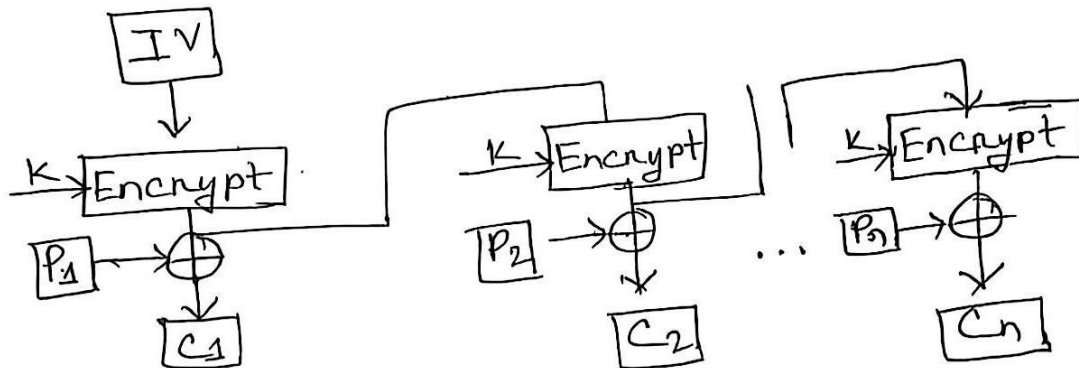
- Since, there is some data loss due to the use of shift register, thus it is difficult for applying cryptanalysis.
- Can handle data streams of any size.

Disadvantage of CFB:

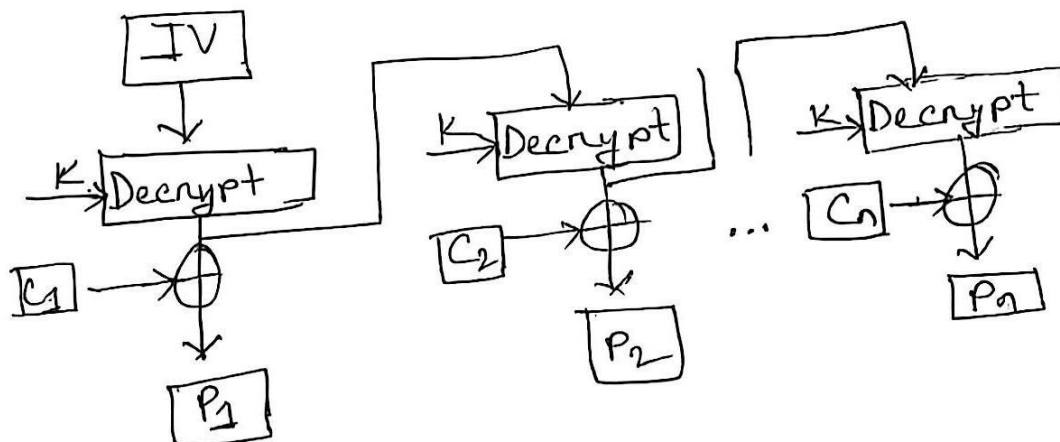
- Slightly more complex and can propagate errors.

OFB Block Diagram:

Encryption



Decryption



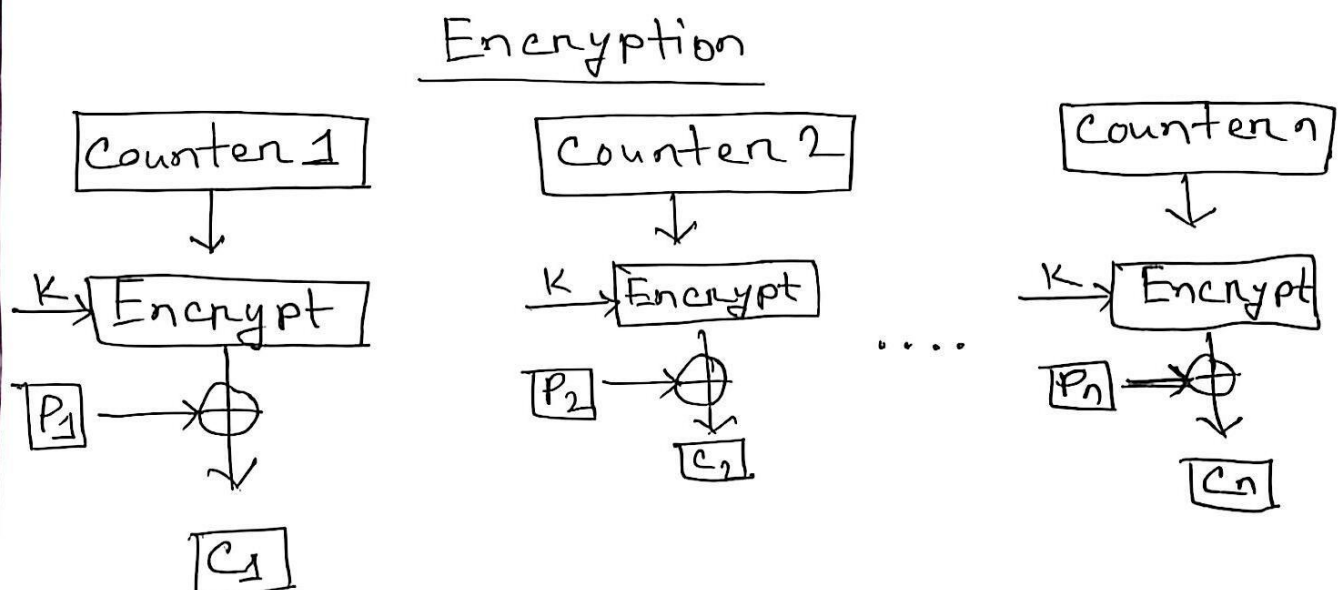
Advantage of OFB:

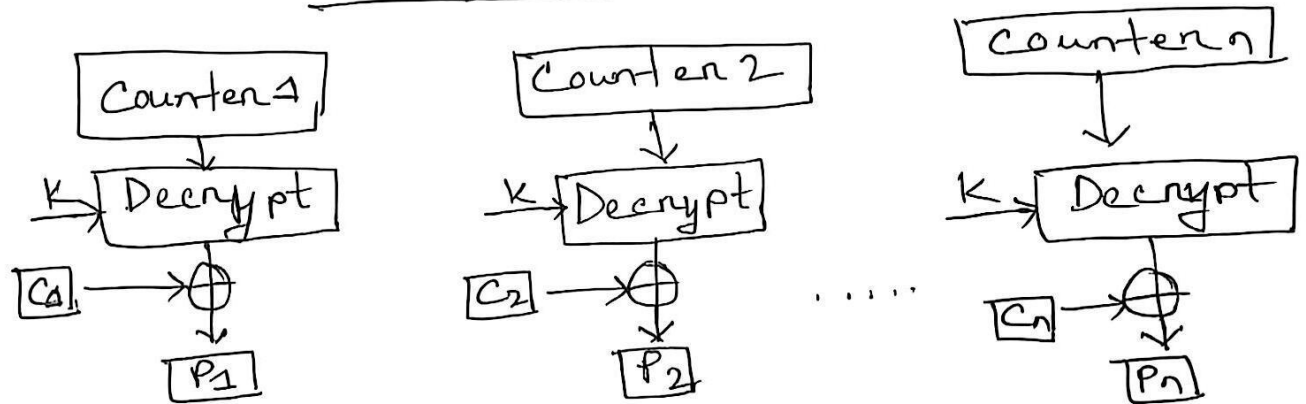
- In the case of OCB, a single bit error in a block is propagated to all subsequent blocks. This problem is solved by OFB as it is free from bit errors.

Disadvantage of OFB:

- It is more susceptible to a message stream modification attack than CFB.
- If the keystream is reused, security is compromised.

Block Diagram of CTR:



DecryptionAdvantage of Counter (CTR):

- Since, there is a different counter value for each block, the direct plaintext and ciphertext relationship is avoided. This means that the same plain text can map to different ciphertext.

Disadvantage of Counter:

- The fact that CTR mode requires a synchronous counter at both the transmitter and the receiver is a severe drawback. The recovery of plaintext is inaccurate when synchronous is lost.

Introduction of RC5:

RC5 is a fast, simple and secure symmetric key block cipher designed by Ron Rivest in 1994.

Key Features:

- Parameterizable:
 - word size (32-bits)
 - Number of round (12)
 - key length (8 bytes)
- Uses:
 - Bitwise operations: XOR, shift, rotate.
 - Modular addition

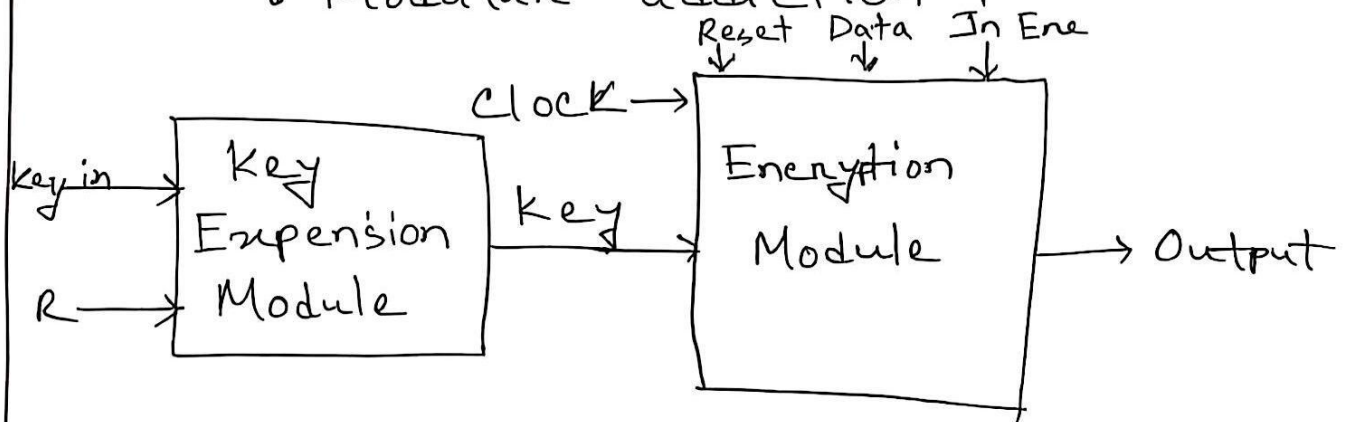
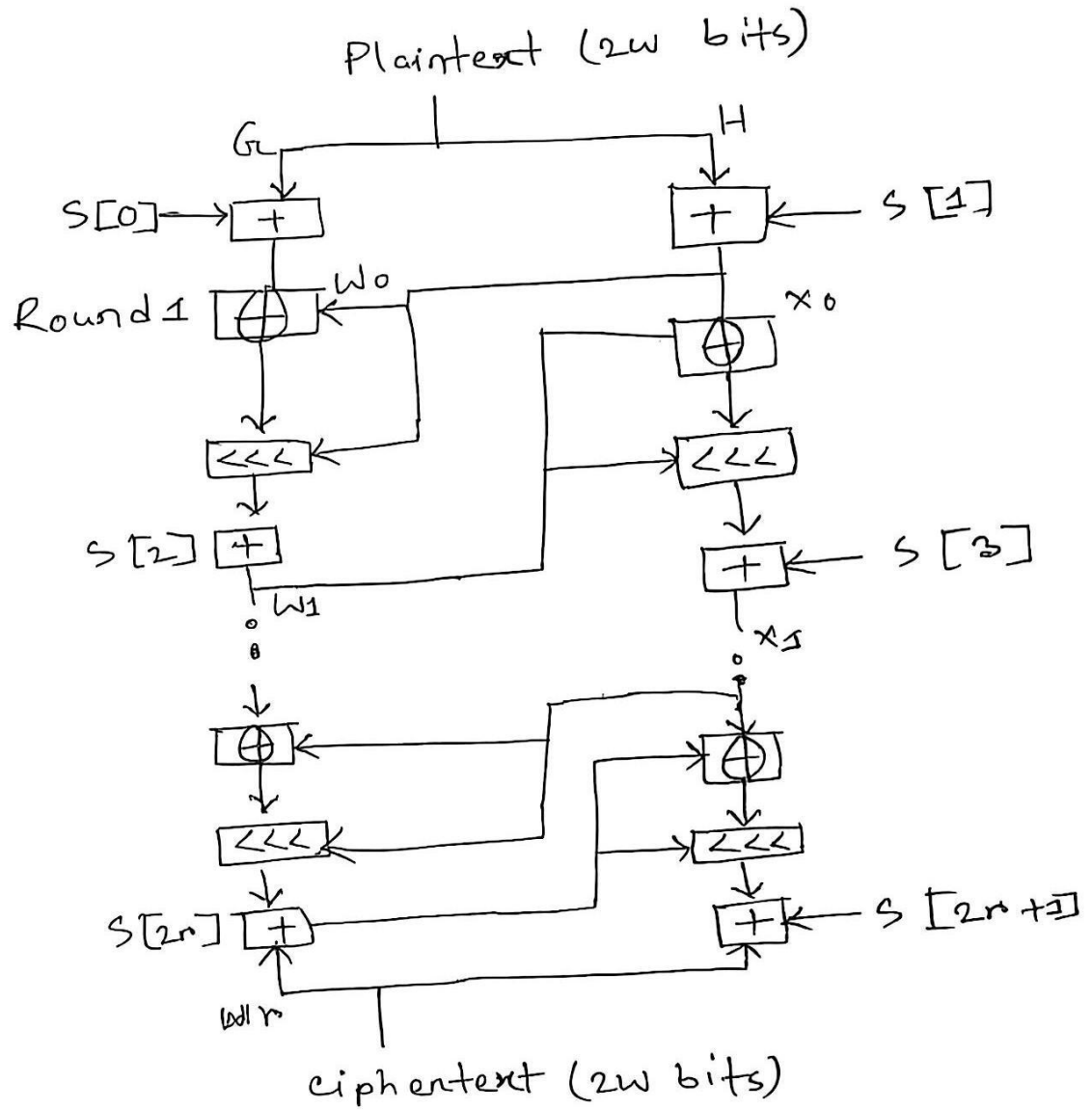


Fig: RC5 Encryption Block Diagram

RC5 Block Diagram:



Java Implementation:

```
Public class RC5 {
```

```
    Private static final int WORDSIZE = 32;
```

```
    Private static final int R = 12;
```

```
    Private static final int B = 8;
```

```
    Private static final int C = B/4;
```

```
    Private static final int T = 2*(R+1);
```

```
    Private int[] S = new int[T];
```

```
    Private static final int P = 0xB7E15163;
```

```
    Private static final int Q = 0x9E3779B9;
```

```
    Public RC5 (byte[] key) {
```

```
        KeySchedule (key);
```

```
    }
```

```
    Private void KeySchedule (byte[] key) {
```

```
        int[] L = new int[C];
```

```
        for (int i = 0; i < B; i++) {
```

```
            L[i/4] = (L[i/4] << B) + (key[i] & 0xFF
```

```
        }
```

$s[0] = P;$

for (int $i=1; i < T; i++$) }

$s[i] = s[i-1] + Q;$

}

int $A=0, B=0, i=0, j=0,$

for (int $k=0, k < 3 * T; k++$) }

$A = s[i] = \text{Integer.rotateLeft}((s[i] + A + B),$

$B = L[j] = \text{Integer.rotateLeft}(L[j] + A + B),$

$i = (i+1) \% T;$

$j = (j+1) \% C;$

}

}

public int T encrypt (int P) }

int $A = P + s[0];$

int $B = P + s[1];$

for (int $i=1; i < R; i++$) }

$A = \text{Integer.rotateLeft}(A \wedge B, A + B) + s[2 * i];$

$B = \text{Integer.rotateLeft}(B \wedge A, A) + s[2 * i];$


```
return new int [] { A, B } ;  
}  
public int [] decrypt (int [] ct) {  
    int B = ct [1];  
    int A = ct [0];  
    for (int i = R; i >= 1; i--) {  
        B = Integer.rotateRight (B - s [2 * i + 1], A) ^ A;  
        A = Integer.rotateRight (A - s [2 * i], B) ^ B;  
    }  
    A = s [0];  
    B = s [1];  
    return new int [] { A, B } ;  
}  
public static void main (String [] args) {  
    byte [] key = "password".getBytes();  
    RC5 rc5 = new RC5 (key);  
    int [] pt = { 0x12345678, 0x9abcde0 };  
    int [] ct = rc5.encrypt (pt);  
}
```


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```
System.out.printf("Encrypted : %08x %08x/n",  
    ct[0]ct[1]);  
  
int[] dt = res.decrypt(ct);  
  
System.out.print("Decrypt : %08x %08x/n",  
    dt[0], dt[1]);  
  
}  
  
}
```

Sample Output :

Encrypted : 7f93d8c2 1423ba29

Decrypted : 12345678 9abcdef0